Covariant Action Candidates and Nonlocal Static Consistency Tests for the Time Attraction Law (Task `50aa2db0`)

Bilateral Top-Hat EOM vs. Screened Helmholtz–Yukawa Action, Static Tangent-Operator Consistency, and Clock Coupling (H0 vs. H1)

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Abstract

In this standalone mathematical and numerical study, I examine Task `50aa2db0`: covariant action candidates and nonlocal static consistency tests for the Time Attraction Law (TAL), building on the locked 3D spherical top-hat formulation (DOI `10.5281/zenodo.23497128`), the dual-arm laboratory program (DOI `10.5281/zenodo.23540741`), the compact-object TOV and radial stability study (DOI `10.5281/zenodo.23602501`), and the V22 Book Edition monograph (DOI `10.5281/zenodo.23604423`). First, I formulate and compare two covariant action closures: **Formulation I** (a nonlocal positive-definite bilateral kernel action with auxiliary fields ϕ_L, λ_L) and **Formulation II** (a local screened Helmholtz–Yukawa scalar-tensor action $(-L^2\Box_g + 1)\phi_L = \alpha_Y \gamma_L \rho_{\rm src}$, $\alpha_Y = 4\pi\alpha_{\rm std}$). Second, I show symbolically that $\nabla_\mu(T^{\mu\nu}_{\rm matter} + T^{\mu\nu}_X) = 0$ holds identically for the algebraically reconstructed X-sector, and I separate this from the numerical residuals (algebraic X-sector quadrature $\lesssim 4 \times 10^{-14}$; full spline divergence $\lesssim 2.7 \times 10^{-5}$); I also solve a flat-kernel fixed-point equilibrium with bilateral reciprocal force (unweighted and lapse-weighted) and an action-consistent Yukawa TOV branch; and I isolate the exact analytic factorization explaining why frozen-f_L pulsation operators ($\delta f_L = 0$) miss the static turning point $dM/d\rho_c = 0$. Third, on the V21 background I run a quasi-dynamic BVP test of the static nonlocal tangent model including the nonlocal perturbation $\delta f_L[\delta\varepsilon, \xi]$, finding along the SLy $L = 10\,$km sequence that the selected BVP-branch eigenvalue crosses zero at $\Delta \log_{10} \rho_{0c} = +6.97\times10^{-5}$ dex (mass mode) and $+7.39 \times 10^{-5}$ dex (gradient mode) from $dM/d\rho_c = 0$, i.e. the nonlocal static tangent operator is consistent with $dM/d\rho_c = 0$; since $\omega^2 = 0$ is set by the static linearization, this zero is invariant under the inertia rescaling $W \to 4W$ (checked numerically), and is not a dynamical proof of the full curved-space window variation, which remains open. Fourth, I show how the choice of Jordan-frame metric in the Dirac–Maxwell matter action selects between observational hypotheses **H0** and **H1** for Sirius B and the 8558 kg granite clock chamber.

1 Covariant Action Formulations: Bilateral Kernel vs. Screened Yukawa

To connect the locked positive-definite exponential law $f_L(x) = \exp(-\phi_L(x)) \in (0, 1]$ (with $\gamma_L = \alpha(G/c^2)L^2$, $\alpha = 1$) to a dynamical variational principle without ad-hoc background bookkeeping, I compare two explicit covariant actions:

$$S[g_{\mu\nu}, \phi_L, \lambda_L, \Psi_{\rm matter}] = \int d^4x\sqrt{-g}\left[\frac{c^4}{16\pi G}R + \mathcal{L}_X(g_{\mu\nu}, \phi_L, \lambda_L) + \mathcal{L}_{\rm matter}(\tilde{g}_{\mu\nu}[\phi_L], \Psi_{\rm matter})\right]. \tag{1}$$

1.1 Formulation I: Nonlocal Positive-Definite Kernel Action and Bilateral Reciprocity

In Formulation I, the auxiliary scalar $\phi_L(x)$ is tied to the spatial top-hat average on the hypersurface Σ_u orthogonal to the fluid four-velocity u^μ ($h_{\mu\nu} = g_{\mu\nu} + u_\mu u_\nu/c^2$) via a Lagrange multiplier field $\lambda_L(x)$:

$$S_{{\rm I},X}[\phi_L, \lambda_L, g_{\mu\nu}] = -\frac{c^4}{8\pi G\gamma_L}\int d^4x\sqrt{-g}\,\lambda_L(x)\left[\phi_L(x) - \gamma_L\int_{\Sigma_u} d^4y\sqrt{-g(y)}\,K_L(x,y)\,\rho_{\rm src}(y)\right], \tag{2}$$

where $K_L(x,y) = \delta(\tau_u(x) - \tau_u(y))\,\Theta(L - d_\Sigma(x,y))/V_L(x)$ is the positive-definite 3D ball kernel of radius L and proper volume $V_L = \frac{4}{3}\pi L^3$. Varying with respect to $\lambda_L(x)$, $\phi_L(x)$, and $\rho_{\rm src}(x)$ yields the Euler–Lagrange closure:

$$\frac{\delta S}{\delta\lambda_L(x)} = 0 \implies \phi_L(x) = \gamma_L\bar{\rho}_L(x), \quad f_L(x) = e^{-\phi_L(x)}; \qquad \frac{\delta S}{\delta\phi_L(x)} = 0 \implies \lambda_L(x) = \frac{8\pi G\gamma_L}{c^4}\frac{\partial\mathcal{L}_{\rm eff}}{\partial\phi_L}(x). \tag{3}$$

Because $\rho_{\rm src}(y)$ appears inside the integral over y, varying the fluid density at x exchanges the dummy integration labels $x \leftrightarrow y$ (*bilateral kernel reciprocity*) and adds a reciprocal potential $V_{\rm recip}(x) = \int d^3y\,K_L(y,x)\lambda_L(y)$ to the fluid Euler equation. In the static spherically symmetric interior, if the auxiliary sector $T^\mu_\nu(X) = {\rm diag}(-\varepsilon_X, P_{r,X}, P_{t,X}, P_{t,X})$ absorbs the anisotropic window stress on the right-hand side of $G^\mu_\nu = \frac{8\pi G}{c^4}(T^\mu_{\nu,{\rm matter}} + T^\mu_{\nu,X})$ while fluid elements follow $\nabla_\mu T^{\mu\nu}_{\rm matter} = 0$ ($P' = -(\varepsilon + P)\nu'$ in geometrized units $G = c = 1$, $\nu = \Phi_{\rm eff}/c^2$, $g_{tt} = -e^{2\nu} = -e^{2\Phi_{\rm eff}/c^2}$), the Bianchi identity $\nabla_\mu G^\mu_r \equiv 0$ uniquely fixes $T^\mu_\nu(X)$ in both V21 modes:

$$\textbf{Mass mode } (m' = 4\pi r^2 f_L\varepsilon,\ \nu' = A_{\rm geo}): \quad \varepsilon_X = -(1 - f_L)\varepsilon, \quad P_{r,X} = 0, \quad P_{t,X} = -\frac{r}{2}(1 - f_L)\varepsilon\nu', \tag{4}$$

$$\textbf{Gradient mode } (m' = 4\pi r^2\varepsilon,\ \nu' = f_L A_{\rm geo}): \quad \varepsilon_X = 0, \quad P_{r,X} = -(1 - f_L)\Big(P + \frac{m}{4\pi r^3}\Big),\ P_{t,X} = P_{r,X} + \frac{r}{2}\big(P'_{r,X} + P_{r,X}\nu'\big), \tag{5}$$

where $A_{\rm geo}(r) = (m + 4\pi r^3 P)/[r(r-2m)]$. Symbolic evaluation in `SymPy` (`conservation_Residuals.json`) confirms $\nabla_\mu T^\mu_r(\text{matter}) \equiv 0$ and $\nabla_\mu T^\mu_r(X) \equiv 0$ identically. In Table 1 and Figure 1(b), evaluating Eqs. (4)–(5) on the numerical TOV grids at $L = 10\,\text{km}$, $\log_{10}\rho_{0c} = 15.0$ gives three distinct tiers (`conservation_Residuals.json`): (i) symbolic identities, residual exactly 0; (ii) algebraic X-sector quadrature residual $\max|\nabla_\mu T^\mu_r(X)|/|\rho_X \nu'| = 4.44\times10^{-16}$ (mass) and $/|P'_{r,X}| = 4.39\times10^{-14}$ (gradient); (iii) full numerically differentiated (spline) background divergence $\max|\nabla_\mu T^\mu_r|/|P'| = 2.70\times10^{-5}$ (mass) and 2.36×10^{-5} (gradient), set by the finite-difference derivative of $h(r)$, not by the action.

Flat-kernel fixed-point equilibrium with bilateral reciprocal force. Eliminating λ_L into the fluid with $\mathcal{L}_{\rm fluid}(n,\phi_L) = -f_L(\phi_L)\,\varepsilon(n)$ and varying ε both explicitly and inside the kernel gives $V_{\rm eff}(x) = f_L(x) - \gamma_L \int K_L(y,x) f_L(y)\varepsilon(y)\,d^3y$ and the modified balance $dh/dr = -A(r) - dV_{\rm recip}/dr$, $V_{\rm recip} = \ln V_{\rm eff}$. I solve this as a self-consistent fixed point (rows `bilateral_eom` in Table 1); it differs from the algebraically closed V21 mass mode. With lapse $N = e^\nu$ in $\int dV\, N f_L \varepsilon$ (kernel and metric held fixed) the variation gives $V_{{\rm eff},N}(x) = f_L - \frac{\gamma_L}{N(x)}\int dV_y\, N(y) K(y,x) f_L \varepsilon$, solved with $P_{\rm matter} = f_L P$ in the lapse equation (rows `bilateral_lapse_weighted`; 2-cell check $N = (1, 0.8)$, $\varepsilon = (1, 2)$, $\gamma = 0.1$: $(0.74882, 0.72084)$ vs. unweighted 0.73160). Full covariant closure still needs the metric variation of the kernel and of the volume element; neither row is an EL solution of action (2).

1.2 Formulation II: Local Screened Scalar-Tensor (Helmholtz–Yukawa) Action

In Formulation II, I replace the nonlocal integral kernel with a local canonical screened scalar action:

$$\mathcal{L}_\phi = -\frac{c^4}{8\pi G\tilde{\alpha}_L}\left[\frac{1}{2}g^{\mu\nu}\nabla_\mu\phi_L\nabla_\nu\phi_L + \frac{1}{2L^2}\phi_L^2\right], \quad \left(-\Box_g + \frac{1}{L^2}\right)\phi_L = \frac{4\pi G}{c^2}\alpha_{\rm std}\,\rho_{\rm src} \iff (-L^2\nabla^2+1)\phi_L = \alpha_Y\gamma_L\rho_{\rm src}. \quad (6)$$

In the static spherical frame, the radial Helmholtz operator $-\frac{1}{r^2}\frac{d}{dr}(r^2\phi'_L) + L^{-2}\phi_L = \frac{4\pi G}{c^2}\alpha_{\rm std}\rho_{\rm src}$ (with $\alpha_Y = 4\pi\alpha_{\rm std}$) has the exact positive-definite Green's function solution

$$\phi_L^{\rm Yuk}(r) = \alpha_Y\,\frac{G}{c^2}\int_0^{R_*} dr'\,\frac{r'L}{r}\,\sinh\frac{r_<}{L}\,e^{-r_>/L}\rho_{\rm src}(r'), \qquad \phi_L^{\rm Yuk}(0) = \alpha_Y\,\frac{G}{c^2}\int_0^{R_*} r' e^{-r'/L}\rho_{\rm src}(r')\,dr', \quad (7)$$

and the variation $\delta S_\phi/\delta g^{\mu\nu}$ yields the local covariant stress-energy tensor $T^{(\phi)}_{\mu\nu} = \frac{c^4}{8\pi G\tilde{\alpha}_L}\big[\nabla_\mu\phi_L\nabla_\nu\phi_L - \frac{1}{2}g_{\mu\nu}\big((\nabla\phi_L)^2 + L^{-2}\phi_L^2\big)\big]$. (`SymPy`: `sym_yukawa_center_limit_residual` $= 0$, no extra factor L). The unit-normalized kernel corresponds to $\alpha_Y = 1$, i.e. $\alpha_{\rm std} = 1/(4\pi) \approx 0.0796$, which reproduces the homogeneous top-hat limit $\phi_L = \gamma_L\rho_{\rm src}$; $\alpha_Y = 1/3$ ($\alpha_{\rm std} = 1/(12\pi)$) is the volume-matched variant. Both normalizations are listed explicitly in `closure_tov_comparison.csv` (`alpha_Y_unit`, `alpha_std_poisson`). Rows `yukawa_flat_phenom_Tphi` are a phenomenological TOV model with flat-space Yukawa field and added scalar stress. Rows `yukawa_action_el` follow from $\mathcal{L} = -[(\nabla\phi)^2 + \phi^2/L^2]/(2\alpha_Y) - f_L\varepsilon$ ($G = c = 1$; `SymPy` residuals `sym_yukawa_el_source_factor` and `_hydrostatic` $= 0$): $B\phi'' + (2B/r + B\nu' + B'/2)\phi' - \phi/L^2 = -\alpha_Y f_L\varepsilon$, $B = 1-2m/r$, $P_{\rm matter} = f_L P$, $h' = -\nu' + \phi'$, $\rho_\phi = (B\phi'^2 + \phi^2/L^2)/(2\alpha_Y)$, matched by $\phi'(R) = -(1/R + 1/L)\phi(R)$; the exterior tail $\Delta M_{\phi,\rm ext}$ is reported separately (SLy, $L = 10$ km: $3.4\times10^{-3}\,M_\odot$ at $\log_{10}\rho_{0c} = 15$, $9.0\times10^{-3}\,M_\odot$ at $M_{\rm max}$). Table 1 and Figure 2 compare Formulation I and Formulation II across SLy, APR4, and H4.

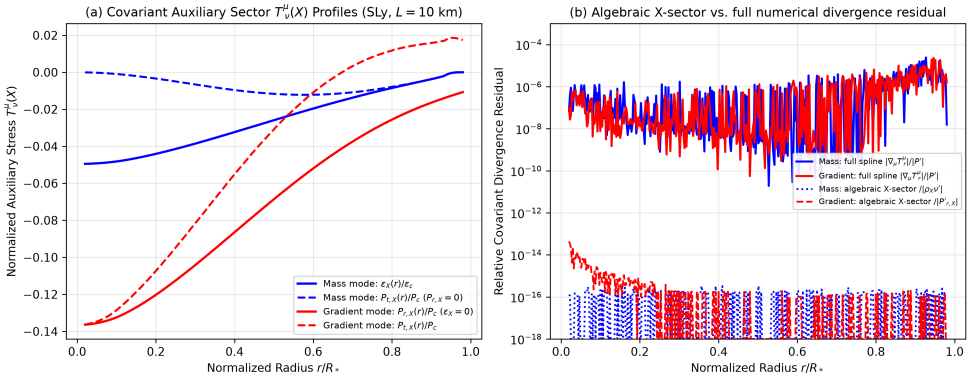


Figure 1: (a) Auxiliary stress-energy components $T^\mu_\nu(X)$ and (b) relative covariant divergence residuals $|\nabla_\mu T^\mu_r|/|P'|$ for SLy at $L = 10\,\text{km}$, $\log_{10}\rho_{0c} = 15.0$.

2 Static Nonlocal Tangent Model and Quasi-Dynamic BVP Test on the V21 Background

In the V21 study (DOI `10.5281/zenodo.23602501`), radial stability was explored using two approximate *frozen-f_L* ($\delta f_L = 0$) Sturm–Liouville operators (`matter` and `metric`). To understand why both frozen-f_L operators miss the static mass peak $dM/d\rho_c = 0$, I first computed in `SymPy` the exact difference between the frozen-f_L Chandrasekhar Sturm–Liouville operator $\mathcal{L}^{\rm matter}_{\rm SL}[\xi]$ and the linearized hydrostatic operator $\mathcal{H}_{\rm lin}[\xi] = (\delta P)' + (\delta\varepsilon + \delta P)\nu' + (\varepsilon + P)\delta\nu'\big|_{\delta f_L = 0}$ in mass mode:

$$\mathcal{L}^{\rm matter}_{\rm SL}[\xi] - \sqrt{\frac{r}{r-2m}}\,e^{2\nu}\,\mathcal{H}_{\rm lin}[\xi]\Big|_{\delta f_L=0} = \frac{4\pi r^{5/2}(f_L-1)\Gamma P\varepsilon e^{2\nu}}{(r-2m)^{5/2}}\Big[(4\pi r^3 P - 3r + 7m)\xi - r(r-2m)\xi'\Big]. \quad (8)$$

Table 1: Comparison of neutron-star TOV sequences under Formulation I (3D spherical top-hat) and Formulation II (screened Helmholtz–Yukawa with $\alpha_Y = 1$ and $\alpha_Y = 1/3$) across SLy, APR4, and H4 from `closure_tov_comparison.csv`.

EOS	L [km]	Mode	Action Kernel	$M_{\max}$ $[M_\odot]$	$R(M_{\max})$ [km]	$\log_{10}\rho_{0c,\mathrm{turn}}$	$R_{1.4}$ [km]	$f_L(0)$ | $f_L(R_*)$ at 10^{15}
SLy	0.0	GR	None ($f_L \equiv 1$)	2.04841	9.9756	15.30242	11.7061	1.000000 | 1.000000
SLy	1.0	mass	Form. I (Top-Hat)	2.04974	9.9785	15.30282	11.7091	0.999169 | 0.999991
SLy	1.0	mass	Form. II (Yukawa $\alpha_Y = 1$)	2.04964	9.9784	15.30278	11.7090	0.999199 | 0.999947
SLy	10.0	mass	Form. I (Top-Hat)	2.09822	10.2029	15.30814	11.8703	0.950477 | 0.987551
SLy	10.0	gradient	Form. I (Top-Hat)	2.10848	10.0182	15.29876	11.7920	0.950680 | 0.987433
SLy	10.0	mass	Form. II (Yukawa $\alpha_Y = 1$)	2.06707	10.0523	15.30566	11.7689	0.983728 | 0.994497
SLy	10.0	gradient	Form. II (Yukawa $\alpha_Y = 1$)	2.07112	9.9952	15.30050	11.7417	0.983762 | 0.994493
SLy	10.0	mass	Form. II (Yukawa $\alpha_Y = 1/3$)	2.05456	10.0008	15.30349	11.7270	0.994576 | 0.998170
SLy	1.0	mass	Bilateral EOM	2.04275	9.9615	15.30412	11.6744	0.999176 | 0.999990
SLy	10.0	mass	Bilateral EOM	1.93214	9.5299	15.37713	10.9416	0.957872 | 0.988372
SLy	10.0	mass	Bilat. lapse-wt.	1.90217	9.5338	15.38661	10.9048	0.958539 | –
SLy	10.0	mass	Yukawa flat phen.	2.06211	10.0210	15.30422	11.7449	0.983754 | 0.994496
SLy	1.0	mass	Yukawa action EL	2.04688	9.9687	15.30316	11.6935	0.999198 | 0.999956
SLy	10.0	mass	Yukawa action EL	2.03583	9.9038	15.31684	11.5456	0.981575 | 0.993556
APR4	0.0	GR	None ($f_L \equiv 1$)	2.18800	9.8997	15.28232	11.3197	1.000000 | 1.000000
APR4	10.0	mass	Form. I (Top-Hat)	2.24152	10.1359	15.28796	11.4757	0.948546 | 0.986533
APR4	10.0	gradient	Form. I (Top-Hat)	2.24877	9.9482	15.27761	11.3937	0.948803 | 0.986420
APR4	10.0	mass	Form. II (Yukawa $\alpha_Y = 1$)	2.20777	9.9824	15.28512	11.3793	0.983526 | 0.994183
APR4	10.0	mass	Bilateral EOM	2.08816	9.6262	15.34020	10.7328	0.956923 | 0.987475
APR4	10.0	mass	Bilat. lapse-wt.	2.05956	9.6246	15.35118	10.7038	0.957732 | –
H4	0.0	GR	None ($f_L \equiv 1$)	2.01385	11.6993	15.20909	13.9522	1.000000 | 1.000000
H4	10.0	mass	Form. I (Top-Hat)	2.05990	11.8833	15.21776	14.0952	0.951154 | 0.989702
H4	10.0	gradient	Form. I (Top-Hat)	2.06963	11.7036	15.21068	14.0267	0.951249 | 0.989545
H4	10.0	mass	Form. II (Yukawa $\alpha_Y = 1$)	2.03060	11.7635	15.21307	14.0096	0.982635 | 0.994798
H4	10.0	mass	Bilateral EOM	1.80990	10.6968	15.33477	12.7249	0.956381 | 0.990180
H4	10.0	mass	Bilat. lapse-wt.	1.78279	10.7415	15.34078	12.6856	0.956941 | –

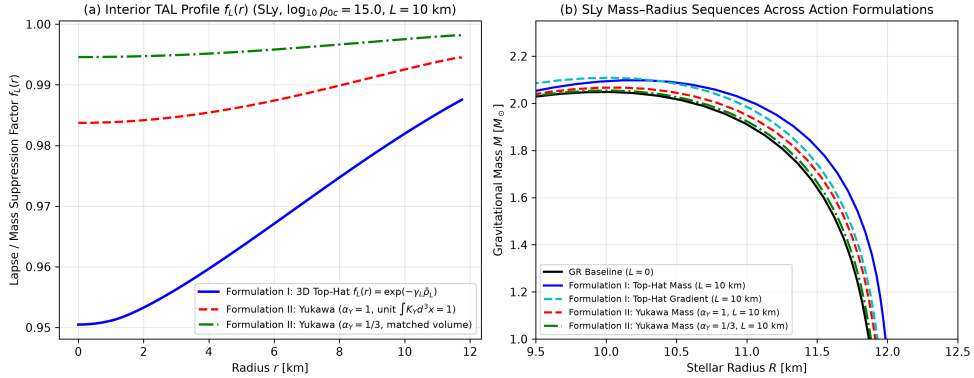


Figure 2: (a) Interior suppression factor $f_L(r)$ at $\log_{10}\rho_{0c} = 15.0$ and (b) SLy mass–radius curves at $L = 10\,\mathrm{km}$ comparing Formulation I (3D top-hat) and Formulation II (screened Yukawa, $\alpha_Y = 1$ and $\alpha_Y = 1/3$).

Equation (8) vanishes identically when $f_L = 1$ (GR), because Chandrasekhar's integration by parts substitutes the GR Einstein identity $\lambda' + 2\nu' = 8\pi r e^{\lambda}(\varepsilon + P)$. When $f_L \neq 1$, however, $\lambda' + 2\nu' = 8\pi r e^{\lambda}(f_L\varepsilon + P)$, producing the $(f_L - 1)$ residual in Eq. (8) AND omitting the nonlocal perturbation $\delta f_L[\delta\varepsilon, \xi]$.

I linearize (around the V21 `dense_profile` background, not the new branches) the coupled system in terms of the enthalpy perturbation $\delta h(r) = \delta P/(\varepsilon + P)$ and mass perturbation $\delta m(r)$ on a radial grid locked to the exact piecewise-polytrope transition radii r_{trans}, including the nonlocal kernel perturbation operator:

$$\delta f_L(r_i) = -\gamma_L f_L(r_i) \sum_j K_{ij} \frac{\varepsilon(r_j) + P(r_j)}{c_s^2(r_j)}\, \delta h(r_j), \qquad \begin{cases} \dfrac{d(\delta h)}{dr} = -\delta\nu'[\delta h, \delta m, \delta f_L[\delta h]] + \omega_0^2\, e^{\lambda - 2\nu}\xi, \\ \dfrac{d(\delta m)}{dr} = 4\pi r^2 \left[f_{\mathrm{mass}} \dfrac{\varepsilon + P}{c_s^2}\delta h + \varepsilon\, \delta f_{\mathrm{mass}}[\delta h] \right]. \end{cases} \tag{9}$$

At $\omega_0^2 = 0$, Eq. (9) is the exact tangent operator to the static equilibrium equations with respect to $\log_{10}\rho_{0c}$. As shown in Table 2 and Figure 3, the coupled problem (9) along the SLy $L = 10\,\mathrm{km}$ sequences has its zero at $\sim 7 \times 10^{-5}$ dex from $dM/d\rho_c = 0$ (equal to the pure-GR discretization baseline 7.12×10^{-5} dex). Because at $\omega_0^2 = 0$ the problem reduces to $\mathcal{H}_{\mathrm{lin}}[\delta h] = 0$, this agreement verifies the *static* nonlocal linearization, not the dynamical inertia: rescaling $W \to 4W$ at $\log_{10}\rho_{0c} = 15.30427$ gives $\omega_{4W}^2/\omega_W^2 = 0.2500$ ($\omega_W^2 = 4.897 \times 10^6\,\mathrm{s}^{-2}$) with identical shooting function $F(0) = 0.023252$, so the zero crossing is inertia-invariant. The raw BVP inertia matrix W is not symmetric; the symmetrized Galerkin form $\mathbf{M}_{\mathrm{sym}} = \mathbf{D}^T \mathrm{diag}(w\,\Delta r)\mathbf{D}$ is symmetric with $\min \mathrm{diag}\, \mathbf{M}_{\mathrm{sym}} = 5.68 \times 10^{-9} > 0$, a Gram-matrix property, not proof of a positive-definite physical kinetic Hessian. The column `omega0_sq_sym_tangent_s2` gives the first-order tangent estimate $\omega^2 \simeq -F(0)/F'(0)$.

Table 2: Radial stability zero-crossing densities ($\omega_0^2 = 0$) vs. the static turning point $dM/d\rho_c = 0$ along the SLy sequences ($L = 0$ GR and $L = 10\,\mathrm{km}$ in `mass` and `gradient` modes) from `coupled_stability_results.csv` and `conservation_Residuals.json`.

Sequence (SLy EOS)	Static Turning Point $\log_{10}\rho_{0c}$ ($dM/d\rho_c = 0$)	Frozen-f_L `matter` $\Delta\log_{10}\rho_{0c}$ [dex]	Frozen-f_L `metric` $\Delta\log_{10}\rho_{0c}$ [dex]	Frozen-f_L BVP $\Delta\log_{10}\rho_{0c}$ [dex]	**Static tangent model (V21 bg)** $\boldsymbol{\Delta\log_{10}\rho_{0c}}$ **[dex]**	Mismatch Reduction Factor (vs. `matter` / BVP)
GR ($L = 0\,\mathrm{km}$)	15.302445	$+2.91 \times 10^{-7}$	$+2.91 \times 10^{-7}$	$+7.12 \times 10^{-5}$	$\mathbf{+7.12 \times 10^{-5}}$	Baseline ($f_L \equiv 1$)
Mass ($L = 10\,\mathrm{km}$)	15.308269	-1.34×10^{-3}	$+1.64 \times 10^{-2}$	-4.06×10^{-3}	$\mathbf{+6.97 \times 10^{-5}}$	**19.2**× / **58.3**× (235.3× vs. `metric`)
Gradient ($L = 10\,\mathrm{km}$)	15.298753	-3.57×10^{-3}	-3.17×10^{-4} *	-4.39×10^{-3}	$\mathbf{+7.39 \times 10^{-5}}$	**48.3**× / **59.3**× (positive weight)

*In gradient mode at $L = 10\,\mathrm{km}$, the approximate frozen-f_L `metric` operator develops an indefinite weight $\mathcal{W}(r) \leq 0$ near the surface; the static tangent model (9) uses a positive Gram weight (not a proof of kinetic positivity).

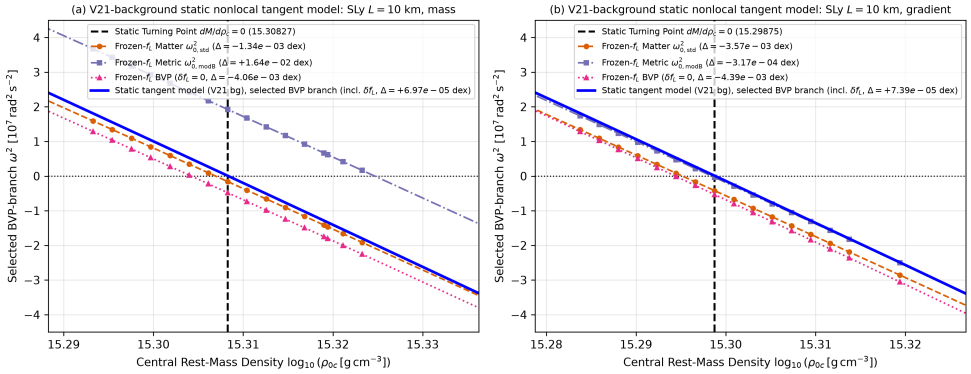


Figure 3: V21-background static nonlocal tangent model: selected BVP-branch eigenvalue $\omega^2(\rho_{0c})$ across the static turning point $dM/d\rho_c = 0$ for SLy at $L = 10\,\mathrm{km}$ in (a) `mass` mode and (b) `gradient` mode, comparing the three uncoupled frozen-f_L operators against the static tangent model (V21 bg).

3 Microscopic Clock and Photon Coupling from the Action (H0 vs. H1)

The covariant action framework makes transparent what determines the choice between observational hypotheses **H0** and **H1** for atomic clocks and stellar spectral lines. Let $\Psi_{\mathrm{atomic}} = (\psi_e, A_\mu)$ denote the Dirac electron and Maxwell electromagnetic fields of an atomic clock at rest with the baryonic fluid four-velocity u^μ, coupled to the effective Jordan-frame metric $\tilde{g}_{\mu\nu}[\phi_L]$:

- **Minimal / Purely Auxiliary Coupling ($\tilde{g}_{\mu\nu} = g_{\mu\nu}$ for $\Psi_{\mathrm{atomic}} \implies$ Hypothesis H0):** If ϕ_L couples only to the bulk baryonic fluid stress-energy tensor while atomic electrons and photons propagate on the physical spacetime metric $g_{\mu\nu}$, the proper time ticked by an atomic clock at $r = R_*$ is $d\tau_{\mathrm{clock}} = \sqrt{-g_{00}(R_*)}\,dt = e^{\Phi_{\mathrm{eff}}(R_*)/c^2}dt$. The surface redshift is purely metric ($1 + z_{\mathrm{H0}} = e^{-\Phi_{\mathrm{eff}}(R_*)/c^2}$), with no direct $f_L(R_*)$ multiplier.
- **Universal Disformal Lapse or Conformal Jordan Coupling ($\tilde{g}_{\mu\nu} = g_{\mu\nu} + \frac{1-f_L^2(\phi_L)}{c^2}u_\mu u_\nu \implies$ Hypothesis H1):** If atomic clocks experience the disformal metric $\tilde{g}_{\mu\nu}$ whose comoving time-time component is $\tilde{g}_{00} = -1 + (1 - f_L^2) = -f_L^2$ for $g_{00} = -1$, $u_0 = -c$ (`SymPy` check `sym_disformal_g00_residual` $= 0$), the proper time increment of an atomic transition is $d\tilde{\tau}_{\mathrm{clock}} = f_L(R_*)\,e^{\Phi_{\mathrm{eff}}(R_*)/c^2}dt$. Photons propagating to infinity on null geodesics conserve coordinate frequency, yielding the direct surface clock factor $1 + z_{\mathrm{H1}} = e^{-\Phi_{\mathrm{eff}}(R_*)/c^2}\,f_L(R_*)^{-1}$.

Table 3 summarizes the resulting numerical predictions from `clock_coupling_H0_H1.csv` for Sirius B ($M = 1.018\,M_\odot$, $R_* = 5586.5\,\mathrm{km}$, $n = 3$ polytrope) and for the V20 8558 kg movable granite chamber ($\rho = 2750\,\mathrm{kg/m^3}$, $V_{\mathrm{tot}} = 3.1120\,\mathrm{m^3}$). Because Sirius B leaves sub-megameter scales $L \lesssim 10^6\,\mathrm{m}$ unconstrained even under H1, both H0 and H1 remain viable at neutron-star ($L \sim 1$–$10\,\mathrm{km}$) and terrestrial laboratory ($L \sim 1\,\mathrm{m}$) scales, with the V20 granite chamber directly testing the H1 clock branch at $|\Delta y_{\mathrm{H1}}| = 1.436 \times 10^{-24}$ ($L = 1.0\,\mathrm{m}$).

Table 3: Quantitative comparison of action-derived clock couplings **H0** ($\tilde{g}_{\mu\nu} = g_{\mu\nu}$) and **H1** (disformal lapse $\tilde{g}_{00} = f_L^2 g_{00}$) for Sirius B ($n = 3$) and the V20 8558 kg granite chamber from `clock_coupling_H0_H1.csv`.

System	L [m]	$f_L(R_*)$ Top-Hat	$f_L(R_*)$ Yukawa	H0 Prediction	H1 Top-Hat Pred.	H1 Yukawa Pred.	Unit
Sirius B ($1.018\,M_\odot$)	1.0×10^3	1.000000	1.000000	80.70055	80.70055	80.70055	km/s
Sirius B ($1.018\,M_\odot$)	1.0×10^4	1.000000	1.000000	80.70055	80.70055	80.70055	km/s
Sirius B ($1.018\,M_\odot$)	1.0×10^6	1.000000	1.000000	80.69839	80.70094	80.74476	km/s
Sirius B ($1.018\,M_\odot$)	5.0×10^6	0.999984	0.999993	80.69487	85.38911	82.85137	km/s
V20 Granite (8558 kg)	0.50	1.000000	1.000000	$+3.075 \times 10^{-48}$	-3.318×10^{-25} (-3.318×10^{-25})	-2.008×10^{-25} (-1.943×10^{-25})	Δy
V20 Granite (8558 kg)	0.85	1.000000	1.000000	$+1.208 \times 10^{-47}$	-1.303×10^{-24} (-1.303×10^{-24})	-3.373×10^{-25} (-3.101×10^{-25})	Δy
V20 Granite (8558 kg)	1.00	1.000000	1.000000	$+1.331 \times 10^{-47}$	-1.436×10^{-24} (-1.436×10^{-24})	-3.781×10^{-25} (-3.407×10^{-25})	Δy

V20 rows: H0 column = TAL excess over Newtonian GR at IN ($\mathbf{r} = \mathbf{0}$); H1 columns = IN shift (IN−OUT differential, $D_{\mathrm{out}} = 2.0\,\mathrm{m}$). Newtonian GR: $\Phi_N(\mathrm{IN})/c^2 = -9.269 \times 10^{-24}$, IN−OUT $= -6.094 \times 10^{-24}$; $\sigma_y = 10^{-18}$ is an illustrative benchmark. Sirius B on the common effective-mass proxy (fixed $n = 3$ Lane–Emden profile with the H0 effective-mass shift, not a newly solved equilibrium) gives at $L = 10^3$ km H0 80.69839, H1 top-hat 80.69878 km/s.

Archival References: D. Szabolcsi, Zenodo DOIs `10.5281/zenodo.23217189` (V14), `10.5281/zenodo.19666220` (V17), `10.5281/zenodo.23497128` (V19), `10.5281/zenodo.23540741` (V20), `10.5281/zenodo.23602501` (V21), and `10.5281/zenodo.23604423` (V22 Book Edition).